INCH-POUND

MIL-C-46793C(ME)
24 April 1992
SUPERSEDING
MIL-C-46793B (ME)
28 October 1983

MILITARY SPECIFICATION

CUPOLA BODY ASSEMBLY FOR ARMORED VEHICLE

BRIDGE LAUNCHER, M60A1 AND M48A5 TANK CHASSIS

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the cupola body assembly required for the armored vehicle bridge launcher, M60A1 and M48A5 tank chassis.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5420

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SPECIFICATION

MILITARY

MIL-T-7O4 - Treatment and Painting of Materiel.

STANDARDS

MILITARY

MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-889 - Dissimilar Metals.

(Unless otherwise indicated, copies of military specifications and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government drawings. The following other Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of solicitation.

DRAWINGS

ME
TA13220E8285 - Cupola Body Assembly.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center ATTN: STRBE-JBS, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
B 46.1 - Surface Texture.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code for Pressure Vessels, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers and Brazing Operations.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

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AMERICAN WELDING SOCIETY (AWS)

D1.1 - Structural Welding Code, Steel.

(Application for copies should be addressed to the American Welding Society, Inc., 550 N.W. LeJeune Road, Miami, FL 33126.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practices for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The cupola body assembly shall be in accordance with TA13220E8285 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly, and operation of the cupola. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified on the applicable drawings.

3.3.1 Material deterioration prevention and control. The item(s) shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environments to which the item may be exposed.

3.3.2 Identification of materials and finishes. The contractor shall

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identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.3 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in the MIL-STD-889.

3.3.4 Cadmium plating. Cadmium plating and cadmium plated items shall not be used.

3.4 Vision blocks. Vision blocks shall be installed in cupolas as specified on drawings. Glass of vision blocks shall be free of sealing compounds and paint after assembly.

3.5 Surfaces. Machine surfaces shall conform to ANSI B 46.1.

3.6 Identification marking. All components shall be marked for identification in accordance with MIL-STD-130.

3.7 Treatment and painting. The portions of the item normally painted internally and externally shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2), topcoat color shall be camouflage green 383. If required, the item shall then be overcoated in accordance with the Government-furnished camouflage patterns with the colors specified in the camouflage patterns.

3.7.1 Government-furnished property (GFP). When specified (see 6.2), the following property in the quantity indicated will be furnished by the Government (see 6.6).

Item No.	Description	Quantity
1	Camouflage paint pattern	As required

3.8 Workmanship.

3.8.1 Castings. All parts, components, and assemblies of the cupola which includes castings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

3.8.2 Metal fabrication. Metal used in fabrication shall be free from kinks or sharp bends. The straightening of material shall be done by methods that shall not cause injury to the material. All bends shall be made with controlled means to ensure uniformity of size and shape. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.8.3 Welding and welders.

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3.8.3.1 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process and shall conform to the provisions of ASME and AWS Codes. All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal all traces of slag shall be removed, and the deposit and adjoining base metal shall be wire brushed clean.

3.8.3.2 Welders or welding operators. Before assigning any welder or welding operator to welding work covered by this specification, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed by either AWS D 1.1 or the ASME code for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined. Contractors who only make horizontal welds need not qualify welders for "all position welding." The contractor is responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes. In the event of evidence of poor welds, the Government reserves the right to require retesting of any welder or welding operator (see 6.5).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspection. Inspection shall be classified as

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follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The cupola body assembly shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each cupola body assembly shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The cupola body assembly shall be examined for the following defects:

101. Loose, damaged, defective, binding, improperly installed or interfering fits of screws, wires, vision blocks, and latch components.
102. Components missing or not as specified (see 3.1).
103. Nonconformance to the dimensions specified on the drawings (see 3.1.1).
104. Material not as specified (see 3.3).
105. Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environments as specified (see 3.3.1)
106. Contractor does not have documentation available for identification of material, material finishes, or treatments (see 3.3.2).
107. Dissimilar metals as specified in MIL-STD-889 are not effectively insulated from each other as specified (see 3.3.3).
108. Cadmium plating/plated items used. (see 3.3.4).
109. Vision of personnel inside of cupola impaired by defective vision blocks or by foreign matter on the vision block surfaces (see 3.4).
110. Surfaces not as specified (see 3.5).
111. Identification marking missing or illegible (see 3.6).
112. Treatment and painting not as specified. Presence of blisters, thin spots, lumps, scratches, or runs (see 3.7).
113. Workmanship not as specified (see 3.8).
114. Castings and metal fabrication not as specified (see 3.8.1 and 3.8.2).
115. Welding and welds not as specified (see 3.8.3.1).
116. Welders and welding operators not certified as required (see 3.8.3.2)

4.6 Inspection of packaging. The preservation, packing and marking shall be examined to determine compliance with the requirements of ASTM D 3951.

5. PACKAGING

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5.1 Preservation, packing and marking (see 6.4). Each complete cupola body assembly shall be preserved and packed in accordance with ASTM D 3951. Marking shall be in accordance with ASTM D 3951. In addition, shipping containers shall display its gross weight and cube.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cupola body assemblies are intended for use as components of vision cupolas for the driver and commander of the armored vehicle bridge launcher, M6OA1 and M48A5 tank chassis.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Time frame required for submission of the first article model, and number required (see 3.2).
d. When camouflage patterns will be provided (see 3.7.1 and 6.6).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Degrees of preservation and packing. Degrees of preservation and packing have not been included as the preservation specified is the minimum acceptable for any level of protection. Marking shall be C/C.

6.5 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title

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a.	3.7.1	DI-MISC-80176	Camouflage line art data
b.	3.8.3.2	DI-MISC-80876	Welding Procedure Qualification Test Report
c.	3.8.3.2	DI-MISC-80875	Welding Procedures

6.6 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.7.1.

6.7 Subject term (key word) listing.

Armored vehicle bridge launcher
Cupola body assembly
Vision cupolas

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Preparing activity:
Army - ME

User interest:
Army - AT

Project 5420-A394

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